Application for a tenure-track position 'junior professor'

1. Curriculum Vitae (max 2 pages)

1.1. Personal information

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1.2. Professional experience

Year	Position	Institution
Oct 2025 - Current	Postdoc	INRIA, Nancy
Mar 2023 - Sep 2025	Postdoc	LAAS-CNRS, Toulouse
Jan 2023 - Feb 2023	Research Engineer	LAAS-CNRS, Toulouse
Jan 2019 - Dec 2022	PhD Student	LAAS-CNRS, Toulouse
Aug 2016 - Nov 2018	Research Assistant	IIT-H, Hyderabad, India

1.3. Scientific experience (maximum 10 lines)

The core of my research is robot motion planning, covering path and trajectory planning, optimal trajectory generation, and high-level control, with a focus on enabling robot autonomy through practical deployment grounded in a strong theoretical foundation. I began by building robots from scratch (stair-climbing robots, a 3D-printed humanoid) at IIIT-Hyderabad, then worked on whole-body motion planning and reinforcement learning for humanoid robots during my Master's. During my PhD and postdoc at LAAS-CNRS, I moved to human-aware motion planning for mobile robots, developing the open-source human-aware navigation stack CoHAN, deployed on PR2, Pepper, TIAGo and other platforms, and used in the euROBIN European project. Since October 2025, I am a postdoc in the HUCEBOT team at INRIA Nancy, working on a new version of CoHAN, robot learning and VLMs, for human-aware navigation and HRI. My expertise lies at the intersection of motion planning and human-robot interaction, with the aim of enabling robots to operate safely and effectively among humans. I have authored 3 journal articles and over 20 conference papers (over 1,000 citations, h-index 13).

1.4. Keywords (maximum 5)

Human-aware motion planning; Social robot navigation; Human-robot interaction; Robot learning; AI for HRI.

1.5. Major events in scientific career

- **2022 - 2026**: Became a recurring organiser of workshops on HRI and social robot navigation at the main robotics conferences, chairing the ICRA 2025 edition, helping to shape the research agenda of the community.
- **2022 - 2026**: Multiple international collaborations on human-aware navigation, including a dual-site, cross-cultural user study with the University of Michigan (USA), leading to HRI 2026 [C21] and ICRA 2026 [C20]; integration of task and motion planning with CNR-ISTC Rome (Italy) [C18]; a research visit to the University of Bremen (Germany) to integrate CoHAN into their Multiverse framework (euROBIN).
- **2024 - 2025**: Authored two field-shaping publications in social robot navigation, a first-author survey and taxonomy in the *International Journal of Robotics Research* [J1] and, as co-author, the community-wide

Principles and Guidelines for Evaluating Social Robot Navigation Algorithms in *ACM Transactions on Human-Robot Interaction* [J2].

- **2025**: Recognised as an expert in the field through appointment as Associate Editor of IEEE Robotics and Automation Letters (RA-L) and co-editor of a Frontiers in Robotics and AI research topic.
- **2025**: Proposal accepted for Dagstuhl Seminar 27131 *Computational Intelligence in Social and Human Aspects of Robot Navigation* (2027), as organiser.

1.6. Grants and other academic activities

- **Grants**: Microsoft Student Travel Grant, International Symposium on Robotics (ISR), Munich, 2018.
- **Editorial**: Associate Editor, IEEE Robotics and Automation Letters (RA-L), Aug 2025 - Aug 2026; Co-editor of the Frontiers in Robotics and AI research topic *Social Robot Navigation - Opportunities, Algorithms, Tools, and Systems* (Mar 2025 - Jul 2026).
- **Organisation**: Organiser, Dagstuhl Seminar 27131 (2027); Organizing Chair, *Advances in Social Navigation: Planning, HRI and Beyond* workshop at the IEEE International Conference on Robotics and Automation (ICRA 2025); Co-organiser of *5th Workshop on Social Robot Navigation: From Humans, to Robots, and to the World* at the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026), *Unsolved Problems in Social Robot Navigation* at Robotics: Science and Systems (RSS 2024), *Social Robot Navigation: Advances and Evaluation* at IROS 2023, and *Joint Action, Adaptation, and Entrainment in HRI* at the ACM/IEEE International Conference on Human-Robot Interaction (HRI 2022).
- **Cascade funding**: Supervised 5 third-party projects from 4 countries, each funded with around €60k through the euROBIN open calls, reviewed their milestone deliverables on which funding releases were based, and wrote the calls for two of them.
  - At LAAS-CNRS, wrote and supervised the calls for *HuNavSim2.0* (Universidad Pablo de Olavide, Spain), funded by the 2nd Open Call of the Technology Exchange Programme in 2024, and *HARD-HAT* (Ingeniarius, Portugal), funded by the 1st Open Call for Collaborative Projects in 2025–2026, which resulted in [J3].
  - At INRIA, supervised *UMI-LINGO* (Eurecat, Spain) and *VILMA* (CYENS, Cyprus), which use a UMI device to collect manipulation data, and *SyntetiQ* (Syntetiq Ltd, UK), which developed a pipeline and collected manipulation data in NVIDIA Isaac Sim. All three were funded by the 3rd Open Call of the Technology Exchange Programme in 2025–2026.
- **Reviewing**: *Journals*: International Journal of Robotics Research (IJRR), IEEE RA-L, International Journal of Social Robotics (IJSR), IEEE Transactions on Robotics (T-RO), ACM Transactions on Human-Robot Interaction (THRI); *Conferences*: ICRA, IROS, HRI, IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), International Conference on Social Robotics (ICSR), IEEE-RAS International Conference on Humanoid Robots (Humanoids), and Iberian Robotics Conference (ROBOT 2026, Program Committee member).
- **Supervision**:
  - Marcela Karam (summer intern, Jun - July 2026)
  - Hariharan Arunachalam (PhD student, LAAS-CNRS, day-to-day supervision as postdoc, supervisor: Dr. Rachid Alami, Aug 2024 - Sep 2025, resulting in [C22])
  - Maxime Nee (summer intern, Jul - Sep 2023)
  - Gerardo Pérez (exchange student, Jul - Sep 2023)
  - Matteo Matera (Master's thesis, co-supervised, Apr - Jul 2023)

1.8. Outreach and popularising science

- **European Parliament**: Presented a robot demonstration as a member of the INRIA team at the AI-powered Robotics event at the European Parliament, covered by the media (September 2026).
- **MuMMER**: Contributed to the navigation system of the Pepper robot deployed as a guide at the Ideapark shopping mall (Tampere, Finland) within the MuMMER European project, interacting with the public [C17].
- **Open source**: My navigation software (CoHAN) and simulation tools are on GitHub (https://github.com/sphanit) and Zenodo [S1], [S2], including an online virtual lab open to anyone through a web browser. The Bi^3 dataset [C20] is also publicly available (https://fluentrobotics.com/bi3dataset/).

2. Research activities

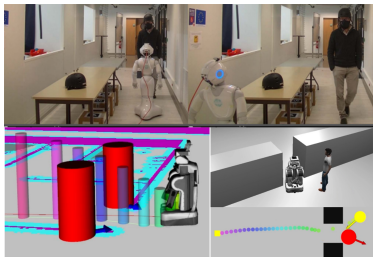
2.1. Description of scientific background

Bachelor's and Master's (2014 - 2018): During my bachelor's and master's at IIIT-Hyderabad, I have worked on several kinds of robotics projects. I joined the Robotics Research Center (RRC) as an honours student and worked on the development of a stair-climbing robot as my first project. It involved using CAD tools like Solidworks to build and analyse the robot, quasi-static phase analysis and prototyping the robot design [C1], [C2]. Following this, I assisted in the development of 3D printed humanoid robot [C3] for building the required circuitry and system integration. In my master's, I explored various aspects of humanoid whole-body motion planning and developed novel solutions to inverse kinematics. I proposed a reinforcement learning-based methodology for finding stable inverse kinematics configurations [C4], which was later extended to dual-arm reachability tasks [C6]. As part of this extension, I introduced a new reinforcement learning architecture, DiGrad [C5]. Further, I completed a three-month exchange at IIT-Jodhpur under Prof. Suril Shah, where I mentored three bachelor's students on project implementation. The project involved capturing human motion using a motion capture system and transferring these motions to a humanoid robot.



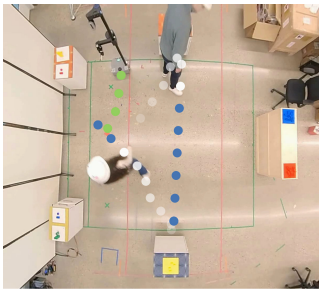
Stair-climbing robot prototype and the 3D printed humanoid doing reachability task [C2], [C4], [C5].

PhD (2019 - 2022): The main objective of my doctoral thesis was to develop human-aware robotic systems and provide open-source implementations. During my PhD at LAAS-CNRS, I extensively worked on a human-aware robot navigation stack [C9], [C10] in ROS and made the system publicly available on GitHub. Over time, I iteratively extended the system with new features and implementations, including a multi-modal planning approach [C10], a novel algorithm for detecting hidden humans [C12], and new optimisation constraints that promote socially aware robot behaviour. The system was tested on multiple robotic platforms both at LAAS (PR2, Pepper and TIAGo) [C10], [C12] and at other laboratories (TIAGo, Hello Robot Stretch2, and a custom robot) [C18]. I also contributed to the development of intelligent human agents [C13], [C16], [C14] in simulation for evaluating the navigation system under diverse scenarios. As part of the MuMMER European project, I contributed to the development of the navigation system [C17] for the Pepper robot during a long-term deployment at Ideapark in Tampere, Finland, and remotely supported project partners in conducting the associated long-term user study. I participated in an international social robot navigation symposium involving researchers from around the world, where we discussed key challenges and emerging research directions. This collaboration resulted in a joint article proposing evaluation guidelines for the field [J2].



CoHAN running on Pepper robot (top) [C9], [C10]. Invisible humans in simulation (bottom left) [C12] and InHuS [C16], [C13].

Research Engineer and Postdoc (2023 - Present): After my PhD, I worked as an engineer for two months before beginning my postdoctoral position in March at LAAS-CNRS. During this time, I focused on formalising and refining a set of novel evaluation metrics [C15] I had proposed at the end of my thesis. These metrics provide a better way to quantify human-aware robot navigation under different circumstances, incorporating elements such as danger and human visibility. During my postdoc at LAAS, I engaged in several collaborations while developing a new version of my human-aware robot navigation stack. This updated system incorporated new technologies, such as behaviour trees, and was made more robust. In parallel, I worked on a survey with several researchers across Europe, which was later published as a journal article [J1]. I initiated a collaboration with the University of Michigan, where we conducted a dual-site user study to investigate how different motion prediction mechanisms affect user perception [C20], [C21]. Another collaboration was conducted with researchers from CNR - Institute of Cognitive Sciences and Technologies, Rome, where we proposed a combined task and motion planning framework [C18], [W6] and tested it in a simulated hospital environment.



Screenshot from the dataset of the user-study [C21], [C20] .

As part of the euROBIN project, I collaborated with the University of Bremen and ISIR-Paris. With Bremen, we integrated my robot navigation stack into their virtual environment, which is publicly accessible online. With ISIR, I explored how combining LLMs and geometric reasoning can improve comfort and legibility in human-aware robot navigation. I moved to INRIA, Nancy, in October 2025, where I continue to work on euROBIN and contributed to INRIA's demo at the AI-powered Robotics event at the European Parliament (September 2026). I am currently exploring AI for human-robot interaction, alongside robot learning and human-aware navigation.

2.2. Scientific project for the junior professor chair (3 pages)

Context and Research Axes

My long-term goal is to integrate robots into everyday life as co-workers and assistants (Fig. 1). This requires motion planning that stays robust to disturbances and failures not only in controlled settings but also in uncontrolled, human-populated environments, where the robot must understand the situation, communicate, react appropriately, and know when to stop. Existing approaches in human-aware motion planning are far from realizing this goal. Learned planners capture complex behaviours but generalise poorly to new robots and contexts, whereas model-based classical planners transfer well across platforms but ignore context and model humans as obstacles following deterministic trajectories. My central hypothesis is that *combining learned interaction and context models with stochastic model-based planning yields human-aware motion that is both adaptive and reliable*, a prerequisite for human-AI shared control. Building on my work in human-aware navigation (HAN), planning and HRI, I will focus on two research axes:



Fig. 1: Target scenarios: a cobot collaborating with a worker on assembly (left), a mobile manipulator helping an elderly person (right).

Axis 1: Robust Human-Aware Motion Planning: How can a robot plan motions that remain safe, legible and efficient when the uncertainty in human behaviour depends on the robot's own actions, and which of these properties can be preserved across contexts and embodiments (mobile robots, cobots and robotic arms, humanoids)?

Axis 2: Learning Interaction Models for Human-AI Shared Control: What representation of an interaction situation generalises across contexts, how can human reactions to robot actions be learned from biased data, and, under uncertain human intent, when should the robot lead, follow, yield, or communicate?

Methodology

Robust and Adaptive Motion Planning (Axis 1): Reinforcement learning (RL) and imitation learning (IL) can learn controllers and behaviours that are difficult to model mathematically, but typically require retraining when transferred to a new platform. Model Predictive Control (MPC) [1] is flexible and transferable, but requires accurate modelling and careful tuning. My previous work CoHAN [C9], [C10], uses an MPC-like framework to plan joint human-robot trajectories in a navigation setting by defining different *social constraints*, but assumes deterministic human motion. I plan to build on it and extend it to stochastic MPC (S-MPC) [2, 3], using *chance constraints* [4] to account for uncertainty in human interactions and *robust constraints* to enforce kinematic feasibility and obstacle avoidance. This raises four scientific obstacles. (i) *Decision-dependent uncertainty*: Humans react to the robot, so the distribution of their motion depends on the plan being optimised, whereas standard chance-constrained formulations assume exogenous uncertainty. My hypothesis is that embedding the action-reaction model of Axis 2 in the co-planning formulation, in the spirit of interaction-aware planning [5], yields plans that are both safer and less conservative than planning against fixed predictions. (ii) *Safety without freezing*: Overestimated uncertainty in dense scenes makes robots stop or detour excessively (the *freezing robot problem* [6]). I will use calibrated uncertainty, e.g., conformal prediction [7], to obtain probabilistic safety guarantees with tight bounds, and study the trade-off between safety, efficiency and social comfort. (iii) *Guarantees with learned components*: RL will learn cost terms that are hard to hand-design [8], vision-language-action models (VLAs) [9] will propose candidate actions as guidance (in my experience, generative models alone do not yield reliable trajectories), and the context model of Axis 2 will update the constraints and objective online (e.g., a hospital corridor vs. a shared assembly cell). The open question is which properties (recursive feasibility, constraint satisfaction) are preserved when learned components reshape the optimisation problem online, which I will address by keeping the S-MPC as a verified safety layer. (iv) *Transfer across embodiments*: I will investigate which social constraints are embodiment-independent and which are specific, moving from mobile bases to manipulators sharing a workspace with humans (handovers, collaborative assembly, waste sorting), where the joint plan covers both robot and human arm motions.

Learning Interaction Models for Shared Control (Axis 2): The S-MPC of Axis 1 is only as good as its model of the human. It needs to know what the situation allows, how humans will react to the robot, and when the robot should take or give up control. Learning such models raises three scientific obstacles. (i) *Representing interaction situations*: Interaction models are typically learned for a single context (e.g., a corridor, a restaurant, an assembly cell) and do not transfer, because there is no common representation of what an interaction situation

is. I hypothesise that a small set of *interaction primitives* (pass, yield, follow, wait, hand over) is shared across contexts, and that context mainly changes their parameters and preconditions. I will represent situations as scene graphs [10] of agents, objects and candidate primitives, and fine-tune vision-language models (VLMs) [11] with LoRA [12, 13] to recognise the context and the viable primitives from multimodal cues (vision, speech, gaze, body pose). Validating this hypothesis would make interaction models transferable instead of context-specific. (ii) *Learning reactions from biased data*: HAN datasets such as MuSoHu, THÖR-Magni and my own Bi^3 dataset [C20] cover few contexts, carry collection biases (e.g., visible teleoperator vs. autonomous robot), and only show how humans reacted to the actions the robot actually took. Predicting the reaction to a different action is therefore a *counterfactual* problem, not only a data-harmonisation one [14]. I will learn a probabilistic action-reaction model (graph neural networks with mixture density outputs [15]) that separates the effect of the robot's action from collection biases, drawing on causal representation learning [16], and use user studies as targeted interventions to validate it. Its output distributions directly define the S-MPC chance constraints of Axis 1. (iii) *Arbitration in shared control*: Under uncertain human intent, when should the robot lead, follow, yield, or communicate to reduce this uncertainty? Existing shared-control arbitration [17] mostly blends a human's teleoperation input with autonomy. I will extend it to co-present collaboration, where both agents act physically, by formulating arbitration as decision-making under uncertainty (POMDPs) [18] in which communication is an action with a cost and an information value. This builds on my work on integrated task and motion planning [C18], [W6], towards proactive decision-making in which the robot decides how much autonomy to take.

HRI Studies and Evaluation (Axes 1 and 2): Both axes rely on understanding interaction. I will study how robots should communicate intent, when non-verbal cues such as motion legibility [19] suffice, and when verbal communication or negotiation is needed. LLMs can generate engaging speech [20] and multimodal LLMs can generate gestures [21], but their outputs are inconsistent across similar situations, so I will study how to constrain them for consistency and legibility. A series of user studies (following EU ethics guidelines) will identify the key features of interaction, whether agents' long-term intentions are needed or local observations suffice, and which constraints are common across contexts and which are context-specific. These findings will directly define the S-MPC constraints. In later years, I will extend this to physical HRI for whole-body human-aware motion planning. In parallel, I will continue my community work on evaluating HAN [J2], [C15] by developing metrics, protocols, and benchmarks, notably through the Dagstuhl Seminar I co-organise.

Work Plan and Risks

Timeline and Indicators

Period	Research objectives	Indicators
Year 1-2	Data harmonisation; probabilistic interaction model; S-MPC extension of CoHAN; first intent-communication user studies; context recognition with VLMs.	Harmonised benchmark released; open-source S-MPC planner; 1 PhD student recruited; ANR JCJC submitted; papers at HRI/ICRA/IROS/RA-L.
Years 3-4	Context-adaptive planning on mobile robots and cobots (shared-workspace collaboration); RL/VLA guidance of S-MPC; multi-context user study; integration with task planning.	Cross-platform demos; multi-site user study; 2nd PhD student; ERC Starting Grant / Horizon proposal; journal articles (IJRR, T-RO, THRI).
Year 5+	Stochastic whole-body human-aware planning; physical HRI; collaborative decision-making.	Whole-body demonstrator; long-term deployment; established research line within the team.

Risks and Mitigation:

1. *Heterogeneous or insufficient data*: If harmonised public datasets are insufficient, I will collect targeted data on the LIRIS Robotics platform, building on my experience with the Bi^3 dual-site study [C20].
2. *Real-time computation*: If decision-dependent S-MPC is too slow for online control, I will approximate it by alternating prediction and planning or by sampling-based optimisation, and run the learned models at a lower rate than the controller. CoHAN already runs in real time with a similar structure.
3. *Interaction primitives do not generalise*: If no shared set of primitives emerges, context-specific models remain usable within the same S-MPC, and identifying which interactions transfer and which do not is itself a contribution to the community.
4. *Ethics and recruitment for user studies*: Studies will follow institutional ethics review, the SICAL team's experience with user studies will support their design and recruitment, and I will use simulation tools for human agents [C13], [C14] for early iterations.

Integration in the Host Team

This project fits the LIMELIGHT chair on *Learning for Human-AI Shared Control* at LIRIS (UMR CNRS 5205). My co-planning approach, in which the robot plans jointly for itself and the human, is itself a form of human-AI shared control. In the IMAGINE team, it will provide the planning and control layer that links the team's expertise in learning, vision and multimodal perception (embodied AI, VLA) to safe, adaptive execution on real robots. With the SICAL team, I will collaborate on user studies, evaluation of interaction and intent communication, and interfaces for shared control. Axis 1 extends naturally from mobile robots to human-robot collaboration with manipulators, which I will validate on the LIRIS Robotics platform in the target applications, from assistive robots in healthcare to industrial assembly and waste sorting. The chair's ANR package (€200k) will fund a first PhD student and experiments, and will seed ANR JCJC and ERC proposals. I will also bring my existing collaborations (LAAS-CNRS, INRIA, University of Michigan, euROBIN Network) to LIRIS.

Dissemination and Impact

Results will be published at the main robotics and HRI venues (HRI, ICRA, IROS, RO-MAN) and journals (IJRR, T-RO, RA-L, THRI), and all software, models and datasets will be released open source. I will disseminate to the broader community through workshops, Dagstuhl seminar, and outreach events. Target applications include assistive and service robots in healthcare and elderly care, where CoHAN is already used [22], and collaborative industrial robots (assembly, waste sorting). The same methods could potentially extend to autonomous vehicles [C8] and to intelligent interfaces that communicate the robot's intent and let humans adjust its autonomy. The project aligns with European priorities on human-centred robotics (Horizon Europe 2026–2027, euRobotics Unified Vision), opening routes to European funding.

References

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3. Teaching activities (maximum 2 pages)

3.1. Teaching experience in higher education

I took up teaching as a part of my Bachelors and Masters in India as well as some time part of my PhD in France. I have tried to put together everything here. The tables list the overview of my teaching activities and the description of courses. All the teaching activities were carried out in English in both countries.

Activities in France

In France, I majorly taught some lab courses and a little bit of tutorials. The courses are done with the Electronics and Computer Science Engineering (GEI) department at INSA, Toulouse.

Status	Year	Establishment	Public	Level	Subject	Volume (in hours)	Nature
Vacataire	2022	INSA, Toulouse	Licence	M1	Real-time Systems / Robot Control	11	TP
DCE	2022	INSA, Toulouse	Licence	M1	Real-time Systems / Robot Control	11	TP
DCE	2021	INSA, Toulouse	Licence	L2	Computer Logic and Hardware	7.33	TP
DCE	2021	INSA, Toulouse	Licence	L2	Computer Logic and Hardware	5	TD
					Total	**34.33 heures**	

The TP sessions in Real-time systems consisted of monitoring and mentoring 20-30 students organised as groups of 2, while they were working on their project. It followed the evaluation based on their performances and a small viva of what they learnt during this project. For the Computer Logic and Hardware course, there were around 14 students for TD and 30 students for TP, where the students worked on their assignments under my supervision. Recently, I also gave a 1 hour lecture at the AIxHRI Summer School, about human-aware robot navigation to master's and PhD students (around 50 students).

Activities in India

In India, I have done extensive teaching including taking few classes, conducting tutorials and labs, setting exam papers for some lab courses and finally evaluating exams and assigning grades. I was a teaching assistant for six semesters and typically the duties require an average of 5 hours per week. These 5 hours include teaching, office hours for clearing doubts, paper corrections and meetings. Each tutorial (TD) is typically 1.5 hours long and a labs is 3 hours long with an average of 10 labs per semester. I have approximately calculated these hours and tabulated them below as well provided the description of the courses.

Status	Year	Establishment	Public	Level	Subject	Volume (in hours)	Nature
Teaching Assistant	2017	IIIT-Hyderabad	Licence	L1	Digital Logic and Processors	$\sim$ 57	TD+TP
Teaching Assistant	2017	IIIT-Hyderabad	Licence + Masters	L3–M2	Introduction to robotics	$\sim$ 21	TD
Teaching Assistant	2016	IIIT-Hyderabad	Licence	L1	Digital Logic and Processors	$\sim$ 57	TD+TP
Teaching Assistant	2016	IIIT-Hyderabad	Licence	L2	Communication Theory - 1	$\sim$ 21	TD
Teaching Assistant	2015	IIIT-Hyderabad	Licence	L1	Digital Logic and Processors	$\sim$ 36	TP
Teaching Assistant	2014	IIIT-Hyderabad	Licence	L2	Embedded Hardware Design	$\sim$ 57	TD+TP
					Total	$\sim$ **249 heures**	

I have taught classes of varying sizes during my time in IIIT-Hyderabad. For the Digital Logic and Processors course, I was monitoring around 20 students in TP and over 50 students in TD. Over the time, I was appointed as the head TA for this course twice and there I was handling around 200 students in both TP and TD, while making sure all the TAs conduct their tutorials and labs well. For the Embedded Hardware Design and Communication Theory-1, I was teaching around 50-60 students sharing the responsibilities with another TA. For the Intro to Robotics course, the strength of the class was around 20. In this course, I have taken a couple of full lectures along with my weekly tutorials.

3.2. Teaching Project

As I am actively involved in motion planning and control of autonomous social robotics, with the background of electronics, I plan to do teaching activities around digital logic, embedded systems, robotics and motion planning, control, probabilistic robotics and localization, programming, decision-making and learning, and human-robot interaction. With growing usage of AI tools by students, my policy would be, to not completely restrict them from using them but to use them to understand or accelerate the work. For instance, I would limit their usage for first- and second-year students, while allowing more flexibility from third-year and Master's students. I believe it is essential for students to first develop a strong foundation in coding and algorithm design independently before relying on AI to enhance their work.

At the level of Bachelors, I can teach digital logic, processor architectures, embedded systems and IoT (Arduino, AVR, Micro-controllers, communication protocols, etc.), hardware design, programming languages (C/C++/Python/Matlab) and data structures, signal processing and probability, introduction to robotics, etc. My teaching style would be a combination of slides with physical class room teaching and I am flexible to conduct online lectures if required. The course structure at the bachelors level would be lectures, assignments and a small project focused on the current state-of-the-art or applications. Depending on the course, I also plan to have labs involved in the courses I teach, which can either be used for the course project or dedicated experiments to get hands-on experience on the concepts taught in class.

At the masters level, I will be focusing more on advanced topics like optimal motion planning and control, imitation and reinforcement learning, human-robot interaction, and evolving technologies for social robotics, AI and autonomous vehicles. I plan to make these courses more engaging by combining the teaching with in-class discussions on the current research papers. As social robotics is more practical, most courses will have projects as a part of course or the entire course could be a project on developing some methodologies. For the discussion sessions, students will be grouped and given papers that they need to read and understand before coming to the class. This works at the masters level as students are more autonomous.

My overall teaching activities aim to look at the real-world problems on social robotics and automation with focus on hands-on experience needed to build expert skills, critical thinking and deeper understanding of the concepts. I am flexible, adaptable, and always approachable when students need extra support or clarification in class.

4. Exhaustive list of contracts and funding obtained in research activities (if applicable)

Not applicable.

5. Comprehensive list of publications, books, patents, talks, poster presentations

5.1. Summary

Number of peer-reviewed articles	3
Number of other publications (proceedings, conference proceedings, book chapters, etc.)	29
Number of patents	0
Number of oral communications	9
Number of poster communications	3
Number of invited seminars	3

5.2. Peer-reviewed articles

[J3] Carlos A. P. Pizzino, **Phani Teja Singamaneni**, Micael S. Couceiro, Rachid Alami. "*A Multimodal Dataset for Socially Aware Navigation of Heavy-Duty Construction Robots*", Frontiers in Robotics and AI, 2026. https://doi.org/10.3389/frobt.2026.1873168.

[J2] Anthony Francis and Claudia Pérez-D'Arpino and Chengshu Li and Fei Xia and Alexandre Alahi and Rachid Alami and Aniket Bera and Abhijat Biswas and Joydeep Biswas and Rohan Chandra and Hao-Tien Lewis Chiang and Michael Everett and Sehoon Ha and Justin Hart and Jonathan P. How and Haresh Karnan and

Tsang-Wei Edward Lee and Luis J. Manso and Reuth Mirsky and Sören Pirk and **Phani Teja Singamaneni** and Peter Stone and Ada V. Taylor and Peter Trautman and Nathan Tsoi and Marynel Vázquez and Xuesu Xiao and Peng Xu and Naoki Yokoyama and Alexander Toshev and Roberto Martín-Martín. "*Principles and Guidelines for Evaluating Social Robot Navigation Algorithms*", ACM Transactions on Human-Robot Interaction, 14, 2, Article 34 (June 2025), 65 pages. https://doi.org/10.1145/3700599.

[J1] **Phani Teja Singamaneni**, Pilar Bachiller-Burgos, Luis J. Manso, Anaís Garrell, Alberto Sanfeliu, Anne Spalanzani, Rachid Alami. "*A Survey on Socially Aware Robot Navigation: Taxonomy and Future Challenges*", The International Journal of Robotics Research, 2024; 43(10), pp.1533-1572. https://doi.org/10.1177/02783649241230562.

5.3. Other publications (proceedings, conference proceedings, book chapters, etc.)

Conferences

[C22] Hariharan Arunachalam, **Phani Teja Singamaneni**, Rachid Alami. "*Validating Virtual Reality for Studying Multimodal Human-Robot Interaction in Socially Aware Robot Navigation*", International Conference on Social Robotics (ICSR), 2026.

[C21] Andrew Stratton, **Phani Teja Singamaneni**, Pranav Goyal, Rachid Alami, Christoforos Mavrogiannis. "*How Human Motion Prediction Quality Shapes Social Robot Navigation Performance in Constrained Spaces*", **ACM/IEEE International Conference on Human-Robot Interaction (HRI)**, 2026, https://doi.org/10.1145/3757279.3788664.

[C20] Andrew Stratton, **Phani Teja Singamaneni**, Pranav Goyal, Rachid Alami, Christoforos Mavrogiannis. "*Bi^3: A Biplatform, Bicultural, Biperson Dataset for Social Robot Navigation*", **International Conference on Robotics and Automation (ICRA)**, 2026, https://doi.org/10.1109/ICRA57385.2026.11695969.

[C19] Antareep Singha, Laksh Nanwani, Mathai Mathew, Samkit Jain, **Phani Teja Singamaneni**, Arun Kumar Singh, K. Madhava Krishna. "*Crowd-FM: Learned Optimal Selection of Conditional Flow Matching-generated Trajectories for Crowd Navigation*", **International Conference on Robotics and Automation (ICRA)**, 2026, https://doi.org/10.1109/ICRA57385.2026.11696381.

[C18] **Phani Teja Singamaneni**, Marcello Tascioni, Alessandro Umbrico, Andrea Orlandini, and Rachid Alami. "*Contextual Social Navigation through Integrated Task and Motion Planning*", International Conference on Social Robotics, ICSR + AI 2024. vol 15563, pp 404–416. https://doi.org/10.1007/978-981-96-3525-2_34.

[C17] Amandine Mayima, Guillaume Sarthou, Guilhem Buisan, **Phani Teja Singamaneni**, Yoan Sallami, Jules Waldhart, Kathleen Belhassein, Aurélie Clodic, Rachid Alami. "*How to make a robot guide?*", International Symposium on Experimental Robotics, ISER 2023. Springer Proceedings in Advanced Robotics, vol 30, pp. 483–494. https://doi.org/10.1007/978-3-031-63596-0_43.

[C16] Anthony Favier, **Phani Teja Singamaneni**, Rachid Alami, "*Challenging Human-Aware Robot Navigation with an Intelligent Human Simulation System*", Social Simulation Conference, ESSA 2023. Springer Proceedings in Complexity, pp 505–518. https://doi.org/10.1007/978-3-031-57785-7_39.

[C15] **Phani Teja Singamaneni**, Anthony Favier, and Rachid Alami. "*Towards Benchmarking Human-Aware Robot Navigation: A New Perspective and Metrics*", **IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)**, 2023, pp. 914-921. https://doi.org/10.1109/RO-MAN57019.2023.10309398.

[C14] Olivier Hauterville, Camino Fernández, **Phani Teja Singamaneni**, Anthony Favier, Vicente Matellán, and Rachid Alami. "*Interactive Social Agents Simulation Tool for Designing Choreographies for Human-Robot-Interaction Research*", ROBOT2022: Fifth Iberian Robotics Conference: Advances in Robotics, ROBOT 2022. Lecture Notes in Networks and Systems, vol 590, pp. 514–527. https://doi.org/10.1007/978-3-031-21062-4_42.

[C13] Anthony Favier, **Phani Teja Singamaneni**, and Rachid Alami. "*An Intelligent Human Avatar to Debug and Challenge Human-aware Robot Navigation Systems*", **ACM/IEEE International Conference on Human-Robot Interaction (HRI)**, 2022, pp. 760-764, https://doi.org/10.1109/HRI53351.2022.9889651.

[C12] **Phani Teja Singamaneni**, Anthony Favier, and Rachid Alami. "*Watch out! There may be a Human. Addressing Invisible Humans in Social Navigation*", **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, 2022, pp. 11344-11351, https://doi.org/10.1109/IROS47612.2022.9982186.

[C11] Jérôme Truc, **Phani Teja Singamaneni**, Daniel Sidobre, Serena Ivaldi, and Rachid Alami. "*Khaos: a*

kinematic human aware optimization-based system for reactive planning of flying-coworker", **International Conference on Robotics and Automation (ICRA)**, 2022, pp. 4764-4770, https://doi.org/10.1109/ICRA46639.2022.9811803.

[C10] **Phani Teja Singamaneni**, Anthony Favier, and Rachid Alami. "*Human-Aware Navigation Planner for Diverse Human-Robot Interaction Contexts*", **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, 2021, pp. 5817-5824, https://doi.org/10.1109/IROS51168.2021.9636613.

[C9] **Phani Teja S** and Rachid Alami. "*Hateb-2: Reactive planning and decision making in human-robot co-navigation*", **IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)**, 2020, pp. 179-186, https://doi.org/10.1109/RO-MAN47096.2020.9223463.

[C8] Raghu Ram Theerthala, AVS Sai Bhargav Kumar, Mithun Babu, **S Phaniteja**, and K. Madhava Krishna. "*Motion planning framework for autonomous vehicles: A time scaled collision cone interleaved model predictive control approach*", IEEE Intelligent Vehicles Symposium (IV), 2019, pp. 1075-1080, https://doi.org/10.1109/IVS.2019.8813823.

[C7] Meha Kaushik, Nirvan Singhania, **Phaniteja S** and K. Madhava Krishna. "*Parameter sharing reinforcement learning architecture for multi agent driving*", In Proceedings of the 2019 4th International Conference on Advances in Robotics (AIR '19). Article 31, pp. 1–7. https://doi.org/10.1145/3352593.3352625.

[C6] **Phaniteja Singamaneni**, Parijat Dewangan, Abhishek Sarkar, and Madhava K. Krishna. "*Learning Multi-Goal Inverse Kinematics in Humanoid Robot*", ISR 2018; 50th International Symposium on Robotics, Munich, Germany, 2018, pp. 1-6.

[C5] **S Phaniteja**, Parijat Dewangan, Pooja Guhan, K. Madhava Krishna, and Abhishek Sarkar. "*Learning dual arm coordinated reachability tasks in a humanoid robot with articulated torso*", **IEEE-RAS 18th International Conference on Humanoid Robots (Humanoids)**, 2018, pp. 1-9, https://doi.org/10.1109/HUMANOIDS.2018.8624979.

[C4] **S Phaniteja**, Parijat Dewangan, Pooja Guhan, Abhishek Sarkar, and K. Madhava Krishna. "*A deep reinforcement learning approach for dynamically stable inverse kinematics of humanoid robots*", IEEE International Conference on Robotics and Biomimetics (ROBIO), 2017, pp. 1818-1823, https://doi.org/10.1109/ROBIO.2017.8324682.

[C3] Divyanshu Goel, **S Phani Teja**, Parijat Dewangan, Suril V. Shah, Abhishek Sarkar, and K. Madhava Krishna. "*Design and development of a humanoid with articulated torso*", International Conference on Robotics and Automation for Humanitarian Applications (RAHA), 2016, pp. 1-5, https://doi.org/10.1109/RAHA.2016.7931878.

[C2] Sri Harsha Turlapati, Mihir Shah, **S Phani Teja**, Avinash Siravuru, and Suril V. Shah. "*Stair climbing using a compliant modular robot*", **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, 2015, pp. 3332-3339, https://doi.org/10.1109/IROS.2015.7353841.

[C1] **S Phani Teja**, Sri Harsha, Avinash Siravuru, Suril V. Shah, and K. Madhava Krishna. "*An improved compliant joint design of a modular robot for descending big obstacles*", In Proceedings of the 2015 Conference on Advances In Robotics (AIR '15). Article 68, pp. 1–6. https://doi.org/10.1145/2783449.2783518.

Workshops and Symposia

[W6] **Phani Teja Singamaneni**, Alessandro Umbrico, Andrea Orlandini, and Rachid Alami. "*Adaptive Robot Navigation through Integrated Task and Motion Planning*", **PlanRob Workshop, International Conference on Automated Planning and Scheduling (ICAPS)**, 2023.

[W5] **Phani Teja Singamaneni**, Alessandro Umbrico, Andrea Orlandini, and Rachid Alami. "*Towards Enhancing Social Navigation through Contextual and Human-related Knowledge*", International Conference on Social Robotics, ALTRUIST Workshop, 2022. https://hdl.handle.net/20.500.14243/515335.

[W4] Olivier Hauterville, Camino Fernández, **Phani Teja Singamaneni**, Anthony Favier, Vicente Matellán, and Rachid Alami. "*IMHuS: Intelligent Multi-Human Simulator*", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022, Workshop on Artificial Intelligence for Social Robots Interacting with Humans in the Real World.

[W3] **Phani Teja Singamaneni**, Anthony Favier, and Rachid Alami. "*Invisible Humans in Human-aware Robot Navigation*", International Conference on Robotics and Automation (ICRA), 2022, Workshop on Social Robot Navigation: Advances and Evaluation.

[W2] Anthony Favier, **Phani Teja Singamaneni**, and Rachid Alami. "*Simulating intelligent human agents for intricate social robot navigation*", Robotics: Science and Systems (RSS), 2021, Workshop on Social Robot Navigation.

[W1] Mary Ellen Foster, Bart Craenen, Amol Deshmukh, Oliver Lemon, Emanuele Bastianelli, Christian Dondrup, Ioannis Papaioannou, Andrea Vanzo, Jean-Marc Odobez, Olivier Canévet, Yuanzhouhan Cao, Weipeng He, Angel Martínez-González, Petr Motlicek, Rémy Siegfried, Rachid Alami, Kathleen Belhassein, Guilhem Buisan, Aurélie Clodic, Amandine Mayima, Yoan Sallami, Guillaume Sarthou, **Phani Teja Singamaneni**, Jules Waldhart, et al. "*MuMMER: Socially Intelligent Human-Robot Interaction in Public Spaces*", AAAI Fall Symposium on Artificial Intelligence for Human-Robot Interaction (AI-HRI), 2019. https://doi.org/10.48550/arXiv.1909.06749.

Book Chapters

[B1] **Phani Teja Singamaneni** (contributing author). "*Personal Robots for Enhanced Quality of Life*", Chapter 7, Springer, ISBN 978-3-032-37619-0. (Forthcoming)

Software

[S2] **Phani Teja Singamaneni**. "*CoHAN-Nav2*", version 0.1.0, Zenodo, 2026. https://doi.org/10.5281/zenodo.18836636.

[S1] **Phani Teja Singamaneni**. "*Cooperative Human Aware Robot Navigation (CoHAN 2.0)*", Zenodo, 2025. https://doi.org/10.5281/zenodo.16918132.

5.4. Patents

Not applicable.

5.5. Oral communications

- [C15] *Towards Benchmarking Human-Aware Robot Navigation: A New Perspective and Metrics*, IEEE RO-MAN 2023.
- [C12] *Watch out! There may be a Human. Addressing Invisible Humans in Social Navigation*, IEEE/RSJ IROS 2022.
- [C10] *Human-Aware Navigation Planner for Diverse Human-Robot Interaction Contexts*, IEEE/RSJ IROS 2021.
- [C9] *Hateb-2: Reactive planning and decision making in human-robot co-navigation*, IEEE RO-MAN 2020.
- [C5] *Learning dual arm coordinated reachability tasks in a humanoid robot with articulated torso*, IEEE-RAS Humanoids 2018.
- [C1] *An improved compliant joint design of a modular robot for descending big obstacles*, AIR 2015.
- [W6] *Adaptive Robot Navigation through Integrated Task and Motion Planning*, PlanRob Workshop, ICAPS 2023.
- [W4] *IMHuS: Intelligent Multi-Human Simulator*, Workshop on AI for Social Robots Interacting with Humans in the Real World, IROS 2022.
- [W3] *Invisible Humans in Human-aware Robot Navigation*, Workshop on Social Robot Navigation: Advances and Evaluation, ICRA 2022.

5.6. Poster communications

- [C21] *How Human Motion Prediction Quality Shapes Social Robot Navigation Performance in Constrained Spaces*, ACM/IEEE HRI 2026.
- [C20] *Bi^3: A Biplatform, Bicultural, Biperson Dataset for Social Robot Navigation*, IEEE ICRA 2026.
- [C8] *Motion planning framework for autonomous vehicles: A time scaled collision cone interleaved model predictive control approach*, IEEE IV 2019.

5.7. Invited seminars

- Invited talk, *Advances and Challenges in Robot Social Navigation* Workshop, European Conference on Mobile Robots (ECMR), Padova, Italy, September 2025.
- Guest lecture, U2IS – ENSTA, Paris, France, May 2026.
- Invited lecture on human-aware robot navigation, AIxHRI Summer School, July 2026.